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| Name: Nguyễn Dương Phúc |
| Student ID: 24521386 |
| Class: CS4323.Q12.CTTT |

# OS
# LAB 2

**CHECKLIST**

**1. Homework**

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Explanation** | ☒ | ☒ | ☒ | ☒ |
| **Your code** | ☒ | ☒ | ☒ | ☒ |
| **Results and discussions** | ☒ | ☒ | ☒ | ☒ |

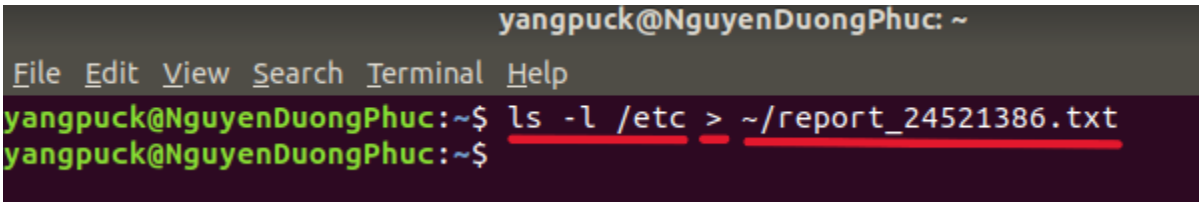
**<u>Mark:</u>** 10/10 (self-scrores)

**Notice: Export PDF file, name format:*
***<MSSV>_LAB2.pdf***

**2. Lab 2**

**1. List the contents of your /etc directory, including the ownership and permissions, and redirect the output to a file called report_<studentID>.txt within your home directory.**

**Answer:**




1. **ls -l /etc**
   - Lists the detailed contents of the **/etc** directory (showing permissions, ownership, size, modification date, file names, etc.).
2. **>**
   - Output redirection operator: writes the command's result into a file instead of printing it on the screen.
3. **~/report_24521386.txt**
   - ~: refers to the current user's **home directory**.
   - The output file will be saved as /home/yangpuck/report_24521386.txt.

```
yangpuck@NguyenDuongPhuc:~$ ls -l /etc
total 1080
drwxr-xr-x  3 root root    4096 Sep 16  2021 acpi
-rw-r--r--  1 root root    3028 Sep 16  2021 adduser.conf
drwxr-xr-x  2 root root    4096 Oct 26 06:50 alternatives
-rw-r--r--  1 root root     401 May 29  2017 anacrontab
-rw-r--r--  1 root root     433 Oct  2  2017 apg.conf
drwxr-xr-x  6 root root    4096 Sep 16  2021 apm
drwxr-xr-x  3 root root    4096 Oct 26 06:49 apparmor
drwxr-xr-x  8 root root    4096 Oct 26 07:06 apparmor.d
drwxr-xr-x  4 root root    4096 Oct 26 06:59 apport
-rw-r--r--  1 root root     769 Apr  4  2018 appstream.conf
drwxr-xr-x  7 root root    4096 Sep 29 09:39 apt
drwxr-xr-x  3 root root    4096 Sep 16  2021 avahi
-rw-r--r--  1 root root    2319 Apr  5  2018 bash.bashrc
-rw-r--r--  1 root root      45 Apr  2  2018 bash_completion
drwxr-xr-x  2 root root    4096 Oct 26 07:04 bash_completion.d
-rw-r--r--  1 root root     367 Jan 27  2016 bindresvport.blacklist
drwxr-xr-x  2 root root    4096 Apr 20  2018 binfmt.d
drwxr-xr-x  2 root root    4096 Oct 26 06:49 bluetooth
-rw-r-----  1 root root      33 Sep 16  2021 brlapi.key
drwxr-xr-x  7 root root    4096 Sep 16  2021 brltty
-rw-r--r--  1 root root   25341 Aug 29  2018 brltty.conf
drwxr-xr-x  3 root root    4096 Sep 16  2021 ca-certificates
-rw-r--r--  1 root root    7817 Oct 26 06:49 ca-certificates.conf
-rw-r--r--  1 root root    6841 Sep 16  2021 ca-certificates.conf.dpkg-old
drwxr-xr-x  2 root root    4096 Sep 16  2021 calendar
```

The file **report_24521386.txt** contains a detailed listing of the **/etc** directory. Each line from **ls -l /etc** includes:

- **Column 1:** File type and permissions (e.g., drwxr-xr-x).
- **Column 2:** Number of hard links.
- **Column 3:** File owner (usually *root*).
- **Column 4:** Group owner (usually *root*).
- **Column 5:** File size in bytes.
- **Columns 6–8:** Last modification date and time.
- **Column 9:** File or directory name.

**2. Sort the contents of the report_<studentID>.txt file from your current directory and append it to the end of a new file named report-sorted.txt.**

**Answer:**

```
yangpuck@NguyenDuongPhuc:~$ sort report_24521386.txt >> report-sorted.txt
yangpuck@NguyenDuongPhuc:~$
```

- **sort**: Sorts the contents of a file alphabetically or numerically based on the text in each line.
- **~/report_24521386.txt**: A file in the user's home directory that contains the unsorted list of files from /etc.
- **>> report-sorted.txt**: Appends the sorted output to report-sorted.txt without overwriting the file.

**After sorted**

```
yangpuck@NguyenDuongPhuc:~$ cat report-sorted.txt
drwxr-s---  2 root dip     4096 Sep 16  2021 chatscripts
drwxr-xr-x 11 root root    4096 Sep 16  2021 X11
drwxr-xr-x  2 root root    4096 Apr 20  2018 binfmt.d
drwxr-xr-x  2 root root    4096 Apr 20  2018 tmpfiles.d
drwxr-xr-x  2 root root    4096 Apr 28  2017 libpaper.d
drwxr-xr-x  2 root root    4096 Aug 24  2021 update-notifier
drwxr-xr-x  2 root root    4096 Feb 24  2018 usb_modeswitch.d
drwxr-xr-x  2 root root    4096 Oct 26 06:49 bluetooth
drwxr-xr-x  2 root root    4096 Oct 26 06:49 skel
drwxr-xr-x  2 root root    4096 Oct 26 06:50 alternatives
drwxr-xr-x  2 root root    4096 Oct 26 06:50 cron.d
drwxr-xr-x  2 root root    4096 Oct 26 06:50 cron.hourly
drwxr-xr-x  2 root root    4096 Oct 26 06:50 cron.monthly
drwxr-xr-x  2 root root    4096 Oct 26 06:50 cron.weekly
drwxr-xr-x  2 root root    4096 Oct 26 06:51 ImageMagick-6
drwxr-xr-x  2 root root    4096 Oct 26 06:51 modprobe.d
drwxr-xr-x  2 root root    4096 Oct 26 06:54 ldap
drwxr-xr-x  2 root root    4096 Oct 26 06:57 terminfo
drwxr-xr-x  2 root root    4096 Oct 26 06:58 profile.d
drwxr-xr-x  2 root root    4096 Oct 26 06:58 sudoers.d
drwxr-xr-x  2 root root    4096 Oct 26 06:59 cron.daily
```

**3. Count the number of words within the report_<studentID>.txt file and *append* the output to the end of a file field2.txt in your home directory. You will need to use both input and output redirection.**

**Answer:**

```
yangpuck@NguyenDuongPhuc:~$ wc -w < ~/report_24521386.txt >> ~/field2.txt
```

- **wc -w**: Counts the number of words in a file.
- **< ~/report_24521386.txt**: Input redirection. Makes wc read the file from standard input, so the output shows **only the word count**, without the file name.
- **>> ~/field2.txt**: Appends the output (word count) to field2.txt in the home directory without overwriting existing content.

```
yangpuck@NguyenDuongPhuc:~$ cat ~/field2.txt
1958
```

⇨ Use **cat ~/field2.txt** to display the result.

- **1958**: The word count of report_24521386.txt

**4. Using the previously created report_<studentID>.txt file, count the number of characters of the last 9 lines.**

**Answer:**

```
yangpuck@NguyenDuongPhuc:~$ tail -n 9 ~/report_24521386.txt | wc -m
531
```

1. **tail -n 9 ~/report_24521386.txt**
   - **tail** displays the last few lines of a file.
   - **-n 9** specifies the number of lines to display (the last 9 lines).
   - **~/report_24521386.txt** is the input file in the home directory.
2. **| (pipe)**
   - Sends the output from tail as input to the next command (wc -m).
   - Allows multiple operations on the data without creating temporary files.
3. **wc -m**
   - **wc** counts lines, words, or characters.
   - **-m** counts the number of characters, including spaces and newline characters, in the input received from the pipe.

⇨ **531** is the total number of characters in the last 9 lines of report_24521386.txt.